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10 CFR 50.73

GNRO2024-00029

August 26, 2024

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2024-003-00,
Feedwater Inlet Check Valve Incorrectly Determined Operable

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report (LER) 2024-003-00, Feedwater Inlet Check Valve Incorrectly Determined Operable. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B), as an operation of condition which was prohibited by the plant's Technical Specifications.

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact me at 802-380-5124.

Sincerely,

A handwritten signature in blue ink, appearing to read 'JAH' followed by a stylized flourish.

JAH/saw

Attachments: Licensee Event Report 2024-003-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

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Washington, DC 20555-0001

GNRO2024-00029

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Attachment
Licensee Event Report 2024-003-00

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-m/doc-collections/nuregs/staff/sr1022/r3/>)

1. Facility Name Grand Gulf Nuclear Station, Unit 1	☒ 050 ☐ 052	2. Docket Number 416	3. Page 1 of 2
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4. Title Feedwater Inlet Check Valve Incorrectly Determined Operable										
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name N/A	☐ 050
04	02	2022	2024	003	00	08	26	2024	Facility Name N/A	☐ 052

9. Operating Mode 4	10. Power Level 0
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(iii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)(i)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ Other (Specify here, in abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Jeff Hard, Regulatory Assurance Manager	Phone Number (Include Area Code) 802-380-5124
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
X	JB	ISV	A585	N					

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year
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16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On April 2, 2022, at 0301 CT, while the plant was in MODE 4 with all control rods inserted during a refuel outage, Grand Gulf Nuclear Station identified a failed post work leakage rate test of Feedwater Inlet Check Valve, 1B21F032B. An operability evaluation of the valve was performed, and the valve was inappropriately determined to be operable. The valve was returned to service, and the plant returned to MODE 1 at the end of the refuel outage.

On June 27, 2024, NRC inspectors identified a Green finding and associated Non-Cited Violation of Technical Specification 3.6.1.3, "Primary Containment Isolation Valves" (PCIV) for failure to ensure each PCIV was operable.

Valve 1B21F032B failed the leakage rate test due to degradation and wear of the valve seat. The operator failed to properly evaluate the operability due to a lack of understanding of the Primary Containment Leakage Rate Testing (Appendix J) program requirements.

There was no impact on the health and safety of the public or reactor safety. The feedwater isolation shutoff valve 1B21F065B would have maintained leakage rates for the containment penetration; therefore, there was no significant risk to containment integrity.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Grand Gulf Nuclear Station, Unit 1	☒ 050	416	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2024	003	00

NARRATIVE**Plant Conditions:**

At the time of the event, Grand Gulf Nuclear Station was in MODE 4 (Cold Shutdown) with all control rods inserted at zero percent reactor power. The station was performing a post-work leakage rate test on Feedwater Inlet Check Valve 1B21F032B.

Description of Event:

On April 2, 2022, during the performance of the post work leakage rate test of the Feedwater Inlet Check Valve 1B21F032B, a failing leakage rate of 7650 milliliters per minute was observed. A Condition Report was issued per procedure, and an operability evaluation was performed using the past operability, the technical specification surveillance requirement, and the Primary Containment Leakage Rate Testing (Appendix J) program requirements. An incorrect assumption about the Appendix J requirements was made, and it was determined that the identified leakage was within the allowable leakage rate based on the surveillance requirements. The operability evaluation incorrectly determined that Feedwater Inlet Check Valve 1B21F032B, was operable, and the valve remained in service for the 24-month fuel cycle.

On June 27, 2024, as part of the Biennial Problem Identification and Resolution Inspection Report 05000416/2024010, the NRC inspectors identified a Green finding and associated Non-Cited Violation of Technical Specification (TS) 3.6.1.3, "Primary Containment Isolation Valves" (PCIV) for failure to ensure each PCIV was operable following a failed leakage rate test of 1B21F032B during a refueling outage. A violation of TS 3.0.4 was also identified for making a mode change without meeting the requirements to do so as a result of not meeting TS 3.6.1.3.

Reportability:

This report is made in accordance with 10 CFR 50.73(a)(2)(i)(B) for an operation of condition which was prohibited by the plant's Technical Specifications. TS LCO 3.6.1.3 requires each PCIV to be OPERABLE. Due to incorrect operability evaluation, Required Actions C and E were not taken.

Cause(s):

The direct cause of the event was degradation and wear of the valve's seat which resulted in the failed leakage rate test. An additional cause of insufficient knowledge of the Appendix J program requirements by the Senior Reactor Operators (SROs) was also identified as the cause of the incorrectly performed operability evaluation.

Corrective Actions:

Completed Actions: Repairs to the valve seat were made during 2024 refuel outage. Post-work leak rate testing was completed satisfactorily and indicated a leak rate of zero millimeters per minute.

Planned Actions: Update Operability Training for SROs with a focus on Code testing requirements.

Safety Significance:

There was no impact on the health and safety of the public or reactor safety. The feedwater isolation shutoff valve 1B21F065B would have maintained leakage rates for the containment penetration; therefore, there was no significant risk to containment integrity.

Previous Similar Events:

LER 2023-001-01 Valve Closure Test Failure Resulting in Condition Prohibited by Technical Specification
LER 2021-001-01 Primary Containment Outer Airlock Door Inoperable Due To Unacceptable Leak Rate